

Revised Load Interconnection System Impact Study Report

For Queue Project Q22-2055

Woodsmere 138 kV
August 16, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2055. The Interconnection Customer proposes a 351.5 MW large load (data center) facility interconnecting at the Woodsmere 138 kV substation in the Virginia area. The request was received on April 05, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 612 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,257,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Ivar / Killian 138kV	Ridgeway / Apline 138kV	213.3	232	100.0
Fieldale / Ivar 138kV line	Apline / Ridgeway 138kV L	176.7	272	100.0
Woodsmere / Fieldae 138kV	Apline / Ridgeway 138kV	167.2	282	100.0
Apline / Ridgeway 138kV	Ivar / Killian 138kV Line	155.1	319	100.0
Ridgeway / Killian 138kV line	Apline Substation / Ridgeway 138kV line	145.9	310	100.0
Woodsmere / Waterford 138kV	Ivar / Killian 138kV	129.1	319	100.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Ivar / Killian 138kV	Rebuild line with higher-capacity conductor	\$7,635,000	85% (\$6,460,000)
Ivar / Fieldale 138kV	Rebuild line with higher-capacity conductor	\$4,069,000	85% (\$3,443,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Woodsmere / Fieldale 138kV	Rebuild line with higher-capacity conductor	\$4,122,000	85% (\$3,488,000)
Apline / Ridgeway 138kV	Rebuild line with higher-capacity conductor	\$3,670,000	85% (\$3,105,000)
Ridgeway / Killian 138kV line	Rebuild line with higher-capacity conductor	\$6,287,000	85% (\$5,320,000)
Woodsmere / Waterford 138kV	Rebuild line with higher-capacity conductor	\$4,441,000	100% (\$4,441,000)
	Total allocated to project		\$26,257,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.